Multiplication of integers

Multiply the moduli, then decide the sign — and two minuses really do make a plus.

LESSON 29–30 | 2 × 40 MIN | MATEMATIKA 7, §12 | S8 1.2

LESSON CLOCK

15' | 25' | 25' | 20' | 5'

The sign rules	15 min
Why two minuses make a plus	25 min
Several factors	25 min
Powers	20 min
Homework	5 min

LEARNING OBJECTIVES

- State and use the sign rules for multiplication.
- Explain why (−a)(−b) = ab.
- Find the sign of a product of several factors.
- Compute powers of negative numbers.

TEXTBOOK

National	pp. 77–82
Cambridge	Stage 8, pp. 6–10
Workbook	Exercise 1.2

01 Explanation

The sign rules

> Multiply the moduli; the product is positive if the signs are alike and negative if they differ

SIGNS	PRODUCT	EXAMPLE
$(+)(+)$	$+$	$4 \times 5 = 20$
$(+)(-)$	$-$	$4 \times (-5) = -20$
$(-)(+)$	$-$	$(-4) \times 5 = -20$
$(-)(-)$	$+$	$(-4) \times (-5) = 20$
anything $\times\ 0$	0	$(-4) \times 0 = 0$

SIGN FIRST, THEN SIZE

Decide the sign from the two signs alone, write it down, then multiply the moduli. Two separate decisions are far more reliable than one combined one.

Why two minuses make a plus

Look at a pattern in which one factor decreases by 1 each time:

PRODUCT	VALUE
$(-4) \times 3$	-12
$(-4) \times 2$	-8
$(-4) \times 1$	-4
$(-4) \times 0$	0
$(-4) \times (-1)$	4
$(-4) \times (-2)$	8

Each step adds 4. Continuing the pattern past zero forces the products to become positive — there is no other way to keep the arithmetic consistent.

THE RULE IS NOT A CONVENTION, IT IS FORCED

If $(-4)(-1)$ were -4, then the distributive law would fail. Mathematics chooses the only definition under which all the old laws keep working.

Several factors

> The product is negative when the number of negative factors is odd, and positive when it is even

PRODUCT	NEGATIVE FACTORS	SIGN	VALUE
$(-2)(-3)(-4)$	3	−	-24
$(-2)(-3)(4)$	2	+	24
$(-1)(-1)(-1)(-1)$	4	+	1
$(-5)(2)(-3)(-1)$	3	−	-30

ONE ZERO FACTOR MAKES THE WHOLE PRODUCT ZERO

However many negatives there are. Scanning for a zero before counting signs saves the whole calculation.

Powers

POWER	MEANING	SIGN	VALUE
$(-3)^2$	$(-3)(-3)$	+	9
$(-3)^3$	$(-3)(-3)(-3)$	−	-27
$(-2)^4$	four factors	+	16
-3^2	$-(3 \times 3)$	−	-9

$(-3)^2$ AND -3^2 ARE DIFFERENT

The brackets decide what is squared. Without them only the 3 is squared and the minus stays outside: $-3^2 = -9$. This distinction is tested in every year from now on.

02 Worked examples

EXAMPLE 1 **Compute $(-4) \times (-5)$, $4 \times (-5)$ and $(-2)(-3)(-4)$.**

1. Alike signs: positive. $4 \times 5 = 20$.
2. Different signs: negative. -20.
3. Three negatives — odd, so negative.
4. $2 \times 3 \times 4 = 24$, so -24.

ANSWER

$20, -20, -24$

EXAMPLE 2 **Compute $(-3)^2$, $(-3)^3$ and -3^2.**

1. $(-3)^2 = (-3)(-3) = 9$

   Two negatives.

2. $(-3)^3 = 9 \times (-3) = -27$

   Three negatives.

3. -3^2 means $-(3^2)$.

   No brackets.

4. $= -9$

ANSWER

$9, -27, -9$

EXAMPLE 3 **Find the sign of $(-5)(2)(-3)(-1)(4)$ without computing it.**

1. Count the negative factors: $-5, -3, -1$.
2. Three of them — an odd number.
3. The product is negative.
4. Its value is -120.

ANSWER

Negative; -120

03 Interactive model

Write the descending pattern (−4)×3, (−4)×2, … on the board and let the class continue it past zero; they produce the rule themselves.

This model could not start: den is not a function

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Multiplication	Ko‘paytirish	Умножение
Product	Ko‘paytma	Произведение
Factor	Ko‘paytuvchi	Множитель
Sign rule	Ishoralar qoidasi	Правило знаков
Power	Daraja	Степень
Even number of factors	Juft sondagi ko‘paytuvchilar	Чётное число множителей
Square	Kvadrat	Квадрат
Cube	Kub	Куб

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

$(-4)(-5)$ equals:

- 20
- -20
- 9
- -9

$4 \times (-5)$ equals:

- 20
- -20
- -1
- 1

A product of three negatives is:

- positive
- negative
- zero
- undefined

$(-3)^2$ equals:

- 9
- -9
- 6
- -6

-3^2 equals:

- 9
- -9
- 6
- -6

A product containing a zero is:

- positive
- negative
- zero
- it depends

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. 4×5

   ANSWER 20

2. $4 \times (-5)$

   ANSWER -20

3. $(-4) \times 5$

   ANSWER -20

4. $(-4) \times (-5)$

   ANSWER 20

5. $(-7) \times 0$

   ANSWER 0

6. $(-3)^2$

   ANSWER 9

7. $(-3)^3$

   ANSWER -27

Medium · the standard question

7 PROBLEMS

1. $(-2)(-3)(-4)$

ANSWER -24

2. $(-2)(-3)(4)$

ANSWER 24

3. $(-1)^4$

ANSWER 1

4. $(-1)^5$

ANSWER -1

5. -3^2

ANSWER -9

6. $(-2)^4$

ANSWER 16

7. $(-5)(2)(-3)(-1)$

ANSWER -30

Hard · stretch and reason

7 PROBLEMS

1. $(-2)^3 \times (-3)^2$

   ANSWER -72

2. $(-1)^{100}$

   ANSWER 1

3. $(-1)^{101}$

   ANSWER -1

4. $(-2)^2 - (-2)^3$

   ANSWER 12

5. The sign of $(-1)(-2)(-3)\dots(-9)$

   ANSWER Negative

6. Find x: $-3x = 21$

   ANSWER -7

7. $(-4)^2 \div (-2)^3$

   ANSWER -2

07 Homework

Homework — 5 tasks

Write the sign of every answer before computing its size.

1. Compute $(-6)(-7)$, $6 \times (-7)$ and $(-6)(7)$.
2. Compute $(-2)(-5)(-3)$ and $(-2)(-5)(3)$.
3. Compute $(-4)^2$, $(-4)^3$ and -4^2.
4. Find the sign of $(-1)(2)(-3)(4)(-5)$ without computing it.
5. Find x if $-5x = 35$.